



COURSE DESCRIPTION FORM: MG-2007 Technology Management

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAMS TO BE EVALUATED

BS-CS – Spring 2026

Course Description

(Fill out the following table for each course in your computer science curriculum. A filled-out form should not be more than 2-3 pages.)

Course Code	MG-2007																								
Course Title	TECHNOLOGY MANAGEMENT																								
Credit Hours	3																								
Prerequisites by Course(s) and Topics																									
Grading Policy	Absolute grading with additive factor																								
Policy about missed assessment items in the course	Compensation of missed assessment item (other than midterm/ final exam) might be held at the end of semester on need basis. For a missed midterm/ final exam, an exam retake/ pre-take application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake/ pre-take cases.																								
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments																								
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	<table><thead><tr><th>Assessment Items</th><th>Number</th><th>Weight (%)</th></tr></thead><tbody><tr><td>Assignments</td><td>2</td><td>5</td></tr><tr><td>Quizzes</td><td>2</td><td>5</td></tr><tr><td>Class Participation</td><td>~</td><td>10</td></tr><tr><td>Sessional</td><td>2</td><td>25</td></tr><tr><td>Project</td><td>1</td><td>15</td></tr><tr><td>Final Exam</td><td>1</td><td>40</td></tr><tr><td>Total</td><td></td><td>100</td></tr></tbody></table>	Assessment Items	Number	Weight (%)	Assignments	2	5	Quizzes	2	5	Class Participation	~	10	Sessional	2	25	Project	1	15	Final Exam	1	40	Total		100
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Course Instructors	Ms Zoya Mashadi																								
Lab Instructors (if any)																									
Course Coordinator																									

URL (if any)	
Current Catalog Description	This course bridges the critical gap between technical expertise and strategic leadership. In today's rapidly evolving digital economy, the ability to manage technical innovation is a key determinant of competitive advantage. This course provides computer science students with the frameworks, concepts, and tools, necessary to lead and manage the process of technological innovation, from ideation to commercialization. The course explores how firms create, commercialize, and capture value from new technologies, focusing on strategies for internal R&D, external technology sourcing, e.g., partnerships, mergers, acquisitions, etc., and building dynamic organizational capabilities. Through a blend of theory, case studies of leading tech firms, such as, Google, Apple, Tesla, SaaS startups etc., and practical project work, students will learn to analyze innovation challenges, design effective development processes, manage project portfolios, and understand the profound ethical and societal implications of the technologies they build. The goal is to equip technically skilled graduates to become strategic contributors and leaders in their organizations.
Textbook (or Laboratory Manual for Laboratory Courses)	• Schilling, M. A. (2024). Strategic Management of Technological Innovation. 7th ed. McGraw-Hill. • Eastery-Smith & Lyles. (2011). Handbook of Organizational Learning and Knowledge Management. • O'Reilly, C. A. & Tushman, M. L. (2021). Lead and Disrupt: How to solve the innovator's dilemma. Stanford Business Books • Milton, N. et al. (2018). Guide to Global Best Practice and Standards in KM. Globe Law and Busienss • Harvard Business Review: selected articles on Agile, Disruption, and Platform Strategies • Case studies: selected case studies on Innovation, Disruption, Platform Strategies, AI Challenges, Talent Management, and Automation
Reference Material	Various HBR and other Case Studies
Course Learning Outcomes	<i>Upon successful completion of this course, students will be able to:</i> 1. Explain the strategic role of technology and innovation management in achieving organizational competitiveness. (2)(2) 2. Analyze the frameworks and processes involved in planning, implementing, reviewing, and controlling innovation initiatives. (4)(3) 3. Evaluate technology adoption models and their application in real-world organizational contexts. (5)(3) 4. Apply appropriate tools and techniques for developing organizational capabilities in managing technology and fostering innovation. (3)(5) 5. Assess the ethical, societal, and legal implications of deploying new technologies and formulate responsible innovation practices. (8)(9)

Topics covered in the course with number of lectures on each topic (assume 15 weeks of instruction and 1.5 hour lecture duration)	Topic Area Number of Weeks Foundation and strategy 3 weeks Internal innovation 4 weeks Obtaining/Sourcing technology (external) 3 weeks Evaluation, Control & Capabilities 3 weeks Ethics, Society & Conclusion 3 weeks			
Laboratory Projects/ Experiments Done in the Course	To be decided/ to be announced			
Programming Assignments Done in the Course	1) Automating some process in your own University - Project 2) Build a business model canvas for a business – Assignment 3) Build a Customer Journey of a Business - Assignment			
Class Time Spent per Week (in percentage)	Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)
	30	30	30	10
Oral and Written Communications	Every student is required to submit at least <u> 5 </u> written reports of typically <u> 5-7 </u> pages and make <u> 1 </u> minor and <u> 1 </u> major oral presentations of typically <u> 2-3 </u> minutes' duration.			

COURSE CONTENTS

Weeks	Contents/ Topics	Courseware Events (Lab/ Case Study/ Quiz/ Assignment/ Project/ Presentation/ Research Report/ Term Paper etc.)	Comments (if any)
Week-01	Introduction to Tech Management & Innovation - The knowledge-driven firm; - Disruptive innovation	Case Study: Kodak vs. Fujifilm	
Week-02	Foundations of Innovation - Applying the S-Curve to current trends, e.g., AI, Quantum 70-20-10 Model of Learning	Mapping learning channels in a tech project	

Week-03	Strategy Formulation - Formulating technology & innovation strategy - SWOT/PESTEL; KM analysis of a tech company as a strategic capability	Assignment: Strategy analysis of a tech firm	
Week-04	Organizing for Innovation - Organizing for innovation: Structures & Teams - Knowledge audits and Mapping	Conduct a mini-knowledge audit for a tech-start up	
Week-05	Development Processes-I Stage-Gate & Waterfall	Process design simulation	
Week-06	Development Processes-II - Agile/Scrum and DevOps - Learning on Demand	Setting up a Jira/Confluence knowledge base	
Week-07	Sessional-I Review Project milestone review		
Week-08	Sourcing Strategies Make, Ally; Buy; Open innovation	Assignment # 2 Acquisition vs. organic growth for knowledge	
Week-09	Strategic Alliances & Ecosystems - Platform strategies - Communities of Practice	Quiz Design a community of practice for an open-source project	
Week-10	Portfolio Management - Roadmapping; - KM Metrics and ROI	Case Study Building a tech portfolio with learning metrics	
Week-11	Sessional – II Leadership and Culture Building a Knowledge-Sharing Culture	Role-play: Leadership responses to knowledge hoarding	
Week-12	Knowledge Capture and Transfer - Tacit vs. explicit knowledge; - AARs, - Storytelling	Activity: Capture a "lessons learned" vignette from a case	
Week-13	Digital KM Tools and AI - KM Systems - AI for search and curation	Quiz AI-powered knowledge base platforms	



Week-14	Ethics, Society & Responsible Innovation • Algorithmic bias • Responsible AI principles	Compensation Assignment Structured debate on an ethics case	
Week-15	KM Strategy & Change Management • Developing a KM strategy • Change management	KM implementation plan for a case	
Week-16	Final Project Presentations and Wrap up • Integration of all concepts		

